

Revised Load Interconnection System Impact Study Report

For Queue Project Q20-1404

Krumkill 100 kV
October 18, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1404. The Interconnection Customer proposes a 136.5 MW large load (data center) facility interconnecting at the Krumkill 100 kV substation in the Illinois area. The request was received on February 12, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 411 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$92,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Wallace Lake – James St 100kV line	Estill Springs 230 kV/1 kV Transformer Bank	106.8	163	9.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wallace Lake – James St 100kV line	Reconductor line	\$389,000	24% (\$92,000)
	Total allocated to project		\$92,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.